Engineering Mechanics 1 · Unit_03 · UNIT 3

DRAFT — UE26ME142A · Unit_03 · UNIT 3 · CLASS 34 Location of the Centroid of a Sector & Semicircular Lamina

Standalone video lesson · authored study notes with source citations

Hugging Face PESU_Materials source tree, official units.json catalogue, transcript sidecars, slide text/PDF/PPT sources and the selected First_Year_8enterprise archive; see the provenance datasets in the release ZIP. Generated 18 October 2026.

· How to use this pack

Read the topic explanation, worked examples, and recap. Evidence citations point to source transcript intervals and slide pages; full slide images remain in the source visual index. Topics include the applicable parts: overview, key terms, formulas, figures, flowcharts, important points, worked examples, past-paper questions and a recap. IDs are shared with the separate documents: E = formula, G = graph, F = flowchart, W = worked example. Each ID is numbered by topic, so E4.2 is the second formula of topic 4.

Source text	Verbatim or OCR-derived text, kept separate from editorial notes.
Candidate	Automated topic/source association requiring manual confirmation.
Low ASR confidence	Transcript segment requires audio verification.
DRAFT	Human review remains incomplete.

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M1

CLASS 34 Location of the Centroid of a Sector & Semicircular Lamina

1 CLASS 34 Location of the Centroid of a Sector & Semicircular Lamina

Study focus: CLASS 34 Location of the Centroid of a Sector & Semicircular Lamina. This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment. EV0000022 · EV0000023

SHORT-NOTE RECAP

Core explanation · 00:00–04:59

Centred of an area of a circular sector.

Key details

- Look at this sector.
- This sector is symmetrical about x axis.
- The purpose for considering this sector symmetrical about x-axis is.
- The center of this sector will be an x-axis, so its y coordinate will be 0.
- In that case, the x coordinate of the centroid will be zero.
- We need to find only the value of y-coordinate of the centroid.
- That's the purpose of considering the figure this way.
- Next, let O be the center of this sector, let capital R be the radius of the sector, then the total angle of this sector, the inscribe the angle let it be 2 alpha.
- Since the total angle is 2 alpha, this angle with respect to x axis will be alpha and the remaining angle will also be alpha.
- Next we need to find only the y coordinate of the.
- Sorry, x coordinate of the centroid.
- To find the x coordinate of the centroid, we have a formula x bar is equal to integral of x into dA divided by total area.
- Which is varying along this radial line.
- It can be anywhere from 0 to capital R.
- Since it's a variable, I'm going to call it as.

- The distance of that element with respect to the center wall, let it be R0.
- Then the radial width is a variable, let it be DR0.
- Angle made by that element with respect to x axis, let it be theta.
- Then the inscriptor the angle of that element let it be d theta.
- D theta will be varying, it's a variable, and theta will be varying from minus alpha to.
- Plus alpha and R0 is varying from 0 to capital R.
- This shaded region is the area which we have considered for the using the formula x bar is equals to integral of x c into d divided by total area.
- Next, what is.
- XC and what is DA in this?
- DA is nothing but area of this element.
- This element is almost like a rectangle.
- What is B here?
- That is the width of the element that is equal to DR0.
- And what is H?
- H is nothing but the arc length that will be equal to R0 into inscribed angle D theta.
- Next, we need to find the value of x is c.
- X c is the centroidal distance of the element with respect to y axis, that x coordinate of the centroid of that smaller element with respect to.
- This is the x coordinate of that element, that itself is x.
- Then how to find this xc?
- When we need to find this XC, we are going to make one assumption that d theta is very small.
- When d theta is very small, theta plus d theta will be approximately equals to theta, and in that case, this element will be just like a.
- Then in that case, how to find X?
- That will be equal to R0 into cos theta.
- We are going to use the cosine rule there.
- Once we know the value of X C and DA, now we should know the value of capital A.
- Capital A is the area of the sector.
- Area of the sector will be equal to half into R squared into inscribed angle, that is to alpha, once we simplify that, we are going to get the value of A as R square into alpha.
- If you don't remember this formula, just integrate this BA.
- And R will be varying from 0 to capital R, and theta is varying from minus alpha to plus alpha.
- If we integrate this by applying the limits, then also we are going to get the value of area of sector.
- Once we know every term in the equation one.
- Substituting the equation one x is r naught cos theta, then d a is r naught d theta into dr0 by capital A.

Example or application from the lecture

- For this, I am going to consider a sector which is symmetrical about.
- And in one more way, you can consider a sector which will be symmetrical about.

- To apply this formula, we have to consider an element within this sector.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000022 · EV0000023 · EV0000024 · +1 source units in pack.json

Core explanation · 05:00–09:59

This apply the limits as I said before R0 is varying from 0 to capital R and R theta is varying from minus alpha to plus alpha.

Key details

- Once we substitute the value of A, we are going to get the X coordinate of the centroid of the sector.
- That will be equal to 2r sin alpha by 3 alpha.
- And to use this formula, you have to remember the alpha should be in terms of.
- radiance and try to understand this formula in this way.
- The distance of the centroid with respect to center on the axis of symmetry that is 2 r sin alpha by.
- Alpha because always the centroid will be on the axis of symmetry for a sector from center to that point centroid on the axis of symmetry is 2 R sine alpha by 3 alpha.
- Again alpha will be radiance and r is the radius of the sector.
- If the semicircle is symmetrical about y-axis, then the x coordinate of the centroid will be 0 because the centroid will be on the axis of symmetry.
- That's the reason why I have considered the figure like this.
- A semicircle which is symmetrical about x-axis, then y-coordinate of the centroid will be 0.
- You need to find only x-coordinate.
- Y bar will be equals to integral of yc into da divided by total area.
- Then how to find yc and dA and total area?
- It can be anywhere from this zero to capital R.
- It is varying in this way.
- Let the inclination of that element with respect to the reference axis, that is x-axis, let it be theta.
- Then the inscribed angle let it be d theta.
- Then the width of the element is also a variable along radial direction.
- That's why we are going to call it as DR naught.
- Next, in this formula, DA stands for area of this element, that is shaded area.
- When d theta is comparatively small, this element will be almost like a rectangle.
- Here B is DR0, that is width of the element.
- Next, we need to find the depth of the element that will be equal to arc length, that is r0 into d theta.
- In this way we can find the value of dA.
- Next, what is yc?
- Yc is nothing but Y coordinate of the centroid of the element.
- In that case, the y distance of that element will be equal to r naught sin theta.
- Because here we are going to get a right angle triangle like this.

- That's why yc will be equal to R0 into sine theta.
- Then substitute the value of BA, then Y C in the equation 1.
- Then capital A is area of the semicircle that is I R square by 2.
- Once we substitute.
- in the equation one, we are going to get it as integral of r naught sin theta.
- into r0 d theta into dr0 divided by total area.
- As I said before, r naught is varying from 0 to capital R and theta is varying from 0 to pi.
- It is a double integration.
- Once we integrate R naught square into with respect to limits from 0 to R it is going to be 2 R cube by 3.
- At the same time integral of sin theta will be cos v.
- Then simplify this again.
- This is the y coordinate of the centroid, and here the centroid will be on the y axis.

Example or application from the lecture

- For this, I am going to consider a semicircle which is symmetrical about y-axis.
- If you consider.
- For this we have to consider one element.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000026 · EV0000027 · EV0000028 · +1 source units in pack.json

Core explanation · 10:00–14:59

Before starting with the numericals, let's know about few basic formulas which we need for solving the numericals in Centroid.

Key details

- For this case, in the first case, where will be the centroid?
- The centroid will be somewhere here.
- Then what is X and Y coordinates of the centroid?
- X bar will be this distance and y bar will be this distance.
- Since the point is in the first quadrant, both x bar and y bar will be positive.
- Look at this, there is a same rectangle of base V and the height, H.
- The centroid is here.
- Since the centroid is in the second quadrant, the x coordinate of the centroid will be negative, and the y coordinate of the centroid will be negative.
- Since the centroid is in the third quadrant, both x and y coordinates of the centroid will be negative.
- That's why it is minus b by two and y by is also minus h by two.
- This is how the values will change depending on the position of the centroid.
- Where will be the centroid here?
- From the derivation, we know centroid will be at a distance of H by 3 with respect to the base of the triangle.
- Then what about with respect to apex of the triangle?

- With respect to apex it will be 2 h by 3.
- And here, since the x-axis is passing through the base of the triangle, y coordinate of the centroid will be h by 3.
- Then what is X bar from the derivation we know X bar is this distance from ear to ear distance.
- That will be equals to base by 3.
- And what is Y bar?
- Y bar is this distance, y coordinate of the centroid.
- That will be h by 3 because x is passing through the base of the triangle.
- And one more, since point is in the first quadrant, both x bar and y bar will be positive.
- Where will be the centroid?
- Then what is X bar?
- X bar is the X coordinate of this C.
- This is x bar.
- Since the point is in the third quadrant, x bar will be negative.
- Then what will be that value?
- With respect to this point, it will be 2b by 3.
- Whereas with respect to this it is B by three.
- Since y axis is passing to this point it will be two b by three minus two b by three.
- Then y bar will be.
- Since the x axis is passing through the ax of the triangle, then the y coordinate of central will be two h by three.
- Since the point is in the third quadrant, y bar is minus two h by three.
- Let B be the base and H be the height.
- Where will be the central?
- Then what is X bar?
- X bar is this distance from here to here.
- What will be that distance?
- 2b by 3 because from this side it is b by 3, from this side it is 2b by 3.
- And at the same time, since the centroid is in the second quadrant, x bar will be negative.
- Next, what is y bar?
- Why by the y coordinate of the centroid?
- That will be with respect to base, since x axis is coinciding with the base, it will be h by three.
- And the point is in the second quadrant, so y coordinate will be positive.
- Next is circle.
- For circle, we know the centroid will be at the center of the circle.
- Then what is X bar?
- X bar is this distance that will be equal to.
- R then what is y bar?
- That is y coordinate of this point C that will be also equal to R.
- And remember point is in first quadrant, so both are positive.

Example or application from the lecture

- The first one is let's consider there is a rectangle of base B and height H.
- Next, let's consider a rectangle.
- Next, let us consider there is a triangle of base B and height H.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000030 · EV0000031 · EV0000032 · +1 source units in pack.json

Core explanation · 15:00–19:59

Of radius R, and where is the center of the circle?

Key details

- That is coinciding with the origin here, both X and Y.
- Then where will be the center here itself?
- If you look at the circle, the circle is symmetrical about this x-axis and this is the center of the circle.
- This is on the x-axis.
- Then what is x bar?
- X bar is this distance that will be equal to radius.
- Since the point is.
- On the negative side of the x axis, x bar will be minus r.
- Then what is y bar?
- Since the point is on the x axis, y bar is equal to 0.
- center here itself, then what is x bar?
- Since the point is on the y axis, x bar will be 0.
- Then what is y bar?
- Y bar is this distance.
- Since it is on the negative side of the y-axis, y bar will be minus.
- And what is this distance?
- Let O be the center of the semicircle.
- Whenever we come across semicircle, remember always we have to start with the axis of symmetry.
- Where will be the axis of symmetry?
- This is the axis of symmetry.
- Then where will be the centroid?
- On the axis of symmetry, somewhere here like this.
- X coordinate of the centroid.
- Since the axis of symmetry and y axis are parallel to each other, x bar will be this distance, that is r.
- And what is y bar?
- We know the distance of the centroid with respect to centre and the axis of symmetry is 4 r by 3 pi.
- Then look at this, the origin as well as the center of the semicircle are coinciding with each other.
- Centroid will be on the axis of symmetry.

- The OC distance is 4R by 3 pi, that we know that is X bar.
- 4 R by 3 pi.
- Since the centroid is on the x axis, y bar is equal to 0.
- Whenever you come across a semicircle, first thing is you have to locate the center of the semicircle.
- This O is the center of the semicircle.
- Then draw the axis of symmetry, then the centroid will be on the axis of symmetry.
- Then the OC distance is 4R by 3 pi, that is from the derivation.
- But what is X bar?
- X bar is this distance from the origin till C.
- Then what is this?
- This will be equal to total that is radius minus O C distance that is 4R by 3 pi.
- And since the point is in the second quadrant, x bar will be negative.
- Next, y bar.
- Y bar, since the point is on the x-axis, y bar will be zero.
- Now, first locate the center of the semicircle.
- This is the center.
- Then draw the axis of symmetry.
- This is the axis of symmetry.
- Next, where will be the center?
- We know from the derivation, this distance OC is 4r by 3 pi.
- But what is X bar?
- This is X bar that is equals to 4R radius.
- Y bar is this distance that will be equals to total R minus 4R by 3 pi.
- Since the point is in the.
- Fourth quadrant y coordinate will be negative.
- Next one is sector.
- We know in this figure.
- The sector is symmetrical about y-axis.
- So x-coordinate of this centroid will be 0 because the centroid is on the y-axis itself.
- And what is y-coordinate of the centroid?
- We know from the derivation of centroid, the distance of the centroid with respect to center on the axis of symmetries.

Example or application from the lecture

- Next consider a circle like this.
- Let's know about the semicircle.
- Let us consider there is a semicircle.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000034 · EV0000035 · EV0000036 · +1 source units in pack.json

Core explanation · 20:00–24:59

Two by third of the R sin alpha by alpha.

Key details

- That's why y bar is this.
- In this case, where will be the centroid?
- On the axis of symmetry.
- Then what is X bar?
- This distance is x bar, then what is that?
- That is equal to 2 r sine alpha by 3 alpha.
- Then what is y bar?
- Since the point is on the x axis, y bar is equal to 0.
- That's it for today's class.

Example or application from the lecture

- Next, consider a sector which is symmetrical about x-axis.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000038 · EV0000039 · EV0000040 · +1 source units in pack.json

A Symbol glossary

Symbol	Name	Meaning	Appears in

B Sources, coverage and gaps

Source	Path	Use
CLASS 34 Location of the Centroid of a Sector & Semicircular Lamina transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/AV_Summaries/034_CLASS_34_Location_of_the_Centroid_of_a_Sector_Semicircular_Lamina__3bf50905d5d6aee5.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
CLASS 34 Location of the Centroid of a Sector & Semicircular Lamina transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/AV_Summaries/034_CLASS_34_Location_of_the_Centroid_of_a_Sector_Semicircular_Lamina__3bf50905d5d6aee5.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Source	Path	Use
CLASS 34 Location of the Centroid of a Sector & Semicircular Lamina transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/AV_Summaries/034_CLASS_34_Location_of_the_Centroid_of_a_Sector_Semicircular_Lamina__3bf50905d5d6aee5.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9984; only the JSON segmentation supplies the displayed timestamps.
CLASS 34 Location of the Centroid of a Sector & Semicircular Lamina transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/AV_Summaries/034_CLASS_34_Location_of_the_Centroid_of_a_Sector_Semicircular_Lamina__3bf50905d5d6aee5.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Historical question-source excerpts are shown only when a page has a strong text-based unit match. They are source references, not new solutions or exam predictions.

Known gaps, conflicts and corrections

1. 15 rendered pages in 033_Centroid__9f0e55503970de0e.slides.md have no readable `.slides.md` text; their original page images are included.
2. 2 rendered pages in 033_UE26ME142A_Unit_3_Class_33_Lecture_PPT__ce948fc3cd6a59bf.slides.md have no readable `.slides.md` text; their original page images are included.
3. 3 rendered pages in 039_UE24CV141A__Unit_3_Beams__9fc34bbd5ca57988.slides.md have no readable `.slides.md` text; their original page images are included.
4. 3 rendered pages in 039_UE25CV141A_Unit_3_Class_48_Lecture_PPT__4d28f283532e284e.slides.md have no readable `.slides.md` text; their original page images are included.
5. 3 rendered pages in 042_UE25CV141A_Unit_3_Class_53_Lecture_PPT__c20af5d61d14aaf0.slides.md have no readable `.slides.md` text; their original page images are included.
6. 5 rendered pages in 043_UE25CV141A_Unit_3_Class_54_Lecture_PPT__c7db1f404daea47d.slides.md have no readable `.slides.md` text; their original page images are included.
7. 6 rendered pages in 044_UE25CV141A_Unit_3_Class_55_Lecture_PPT__741ad989a632bc04.slides.md have no readable `.slides.md` text; their original page images are included.
8. 4 rendered pages in 044_UE26ME142A_Unit_3_Class_44.1_Lecture_PPT__068ee897e4f8117e.slides.md have no readable `.slides.md` text; their original page images are included.

9. No historical archive question page met the unit-match candidate threshold; all subject-level and ambiguous pages remain in datasets/question-archive.json.
10. 6 video-to-slide associations were not promoted to a match. Candidate rankings are listed in the transcript corpus and video-slide-matches.csv; the slide decks appear separately.
11. 1 transcript segments have low ASR confidence; they are flagged with timestamps in the transcript blocks and indexed in the corpus.
12. This is a visibly labelled source-preservation draft. Human page-level semantic and visual review is not complete; no unsupported subject summary or formula correction is claimed.
13. The 19.26 GiB MP4 files remain hosted on Hugging Face; the full video path/hash and transcript coverage are indexed. The full audio/video was not watched end-to-end.
14. Formula candidates are machine-extracted text with source locators; use the adjacent original slide page and the formula index before treating a candidate as an authoritative equation.
15. The study PDF contains edited, topic-organized lecture notes and textual slide-page takeaways. Full-page slide images remain in the source/visual index datasets, not embedded as the lesson. Human content, ASR, formula, and page-level visual review remains pending.

Coverage

Evidence is summarized by source to keep the reading copy legible. Full source-unit IDs, locators and decisions remain in pack.json.

Source	Unit counts	Example decisions
SRC00006	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0022. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0023 to s0043. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0044 to s0063. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00007	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0022. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0023 to s0043. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0044 to s0063. Sidecar variants compare...
SRC00008	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0022. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0023 to s0043. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0044 to s0063. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00009	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0022. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0023 to s0043. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0044 to s0063. Sidecar variants compare...

C Revision and retrieval practice